INCH-POUND

MIL-S-70682(AR)
AMENDMENT 2
20 April 1998
SUPERSEDING
AMMENDMENT 1
26 March 1993

MILITARY SPECIFICATION

SIMULATOR, FLASH, ARTILLERY, M21
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of MIL-S-70682 (AR), dated 30 March 1990, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.4.6 Delete in its entirety.

PAGE 5

4.3.1 Change the First Article quantity column of the following parts to read:

Part Description	Drawing	Quantity
Cap, protecting	9342132	3/cavity
Container, Flash Charge	9342142	3/cavity
Body	9342143	3/cavity
Tube Assembly	9342144	5
Tube	9342145	5
Cover, Flash Charge	9342149	3/cavity
* O-Ring, Cover	9342149	25
Plug, Electrical Assembly	9342154	3/cavity
Insert, Threaded	9342155	5
Pin, Electrical	9342156	5

Under table, line seven, delete "(less quickmatch)".

Add to table:

Plug, Electrical Assembly (Alternate)	1294400	15

4.3.1. Submission. Delete from list: Cap, Short Circuit Assembly in its entirety.

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PAGES 6-8

* TABLE I: Delete in their entirety and substitute new pages included.

PAGE. 9

TABLE. I: Heat Soak: Delete "Heat soak, 108, –, 3.4.4, 4.5.3.4”.

Match and Cable Assembly. Under Electrical resistance test, delete “(less quickmatch)”.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Flash, Artillery M21 and Components	SHEET 1 OF 5	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY ------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cap, Protecting (Dwg. 9342132) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.1
	Container, Flash Charge (Dwg. 9342142) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.2
	Tube (Dwg. 9342145) Examination for defects Moisture content	5, 0-1 (a)	3.2 3.3.2	4.4.2.3 4.5.2.1
	Tube Assembly (Dwg. 9342144) Examination for defects	5, 0-1	3.2	4.4.2.5
	Body (Dwg. 9342143) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.6
	Cover, Flash Charge (Dwg. 9342149) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.8

NOTES:

(a) Sample for this test shall be taken from test samples.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Flash, Artillery M21 and Components	SHEET 2 OF 5	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY ------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	O-Ring (Dwg. 12982986) Examination for defects	25	3.2	4.4.2.8
	Pin, Electrical (Dwg. 9342156) Examination for defects	5, 0-1	3.2	4.4.2.10
	Insert, Threaded (Dwg. 9342155) Examination for defects	5, 0-1	3.2	4.4.2.11
	Pl-ug, Electrical Assembly (Dwg. 9342154) Examination for defects	3/, 0-1 cavity	3.2	4.4.2.12
	Igniter, Electrical Assembly (Dwg. 9342148) Examination for defects Electrical resistance test Functional timing sequence test	15, 0-1	3.2 3.2 3.4.6	4.4.2.15 4.4.2.15 4.5.3.6

NOTES:

4 (7)

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Flash, Artillery M21 and Components	SHEET 3 OF 5	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY ------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Packing and Marking For Box, Packing (Dwg. 9342972) Examination for defects	1 box, 0-1 162 items	3.2	4.4.2.24
	Plug, Electrical Assembly (Alternate) (Dwg. 1294400) Examination for defects	15, 0-1	3.2	4.4.2.25

NOTES:

5 (8)

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PAGE 13

4.4.2.2 Delete minor defect 201 in its entirety.

PAGE 15

4.4.2.4: Delete in its entirety.

PAGE 16

4.4.2.5, Major 101: Delete “Gage” and substitute “Visual - straight edge".

Delete Minor 201 in its entirety

PAGE 17

4.4.2.6 Delete major defects 101, 102 and minor defects 202, 204 in their entirety.

PAGE 18

4.4.2.7: Case, Cartridge Assembly: Add major defect "10l Pull test, Tube Assembly to Body, 100%, –, 3.2, Gage'.

PAGE 19

* 4.4.2.8 Cover, Flash Charge:

Revise Title to “Cover Assembly, Flash Charge”

Change major defect l02 to read;
“102 Slot for cable missing .40% 3.2 Visual"

Delete Major 103 in its entirety.

* Add “Major 106, O-Ring inside diameter, 1.00%, 3.2, Gage”.

* Add “Major 107, O-Ring cross section diameter, 1.00%, 3.2, Gage”.

* Add “Major 108, O-Ring missing or in incorrect location, 100%, 3.2, visual”.

PAGE 20

4.4.2.9: Quickmatch. Delete in its entirety.

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PAGE 21

4.4.2.10 Change major defects 10l, l02, and 103 and minor defect 203 to read:

“101 Length from slotted end to flange, near side.
102 Length from slotted end to flange, far side.
103 O.D. of slotted end before cutting slot.
203 Presence of slot .65% 3.2 Visual".

Add:

“106 Diameter of pin head, –, 0.40%, 3.2, Gage
107 Length of pin head, min, –, 0.40%, 3.2, Gage
108 Width of slots, –, 0.40%, 3.2, Gage”

PAGE 22

4.4.2.11 Add “Note 1” callout following “Major” and “Minor”

Add note:

"1 – These classified defects do not apply to the alternative part which is a commercial item.”

PAGE 23

4.4.2.12 Delete major defects 101 and 104 in their entirety.

Change major defect 102 and 103 to read:
“102 Width under head (Datum B)”.
“103 Length under head (Datum A)”.

PAGE 24

* 4.4.2.13: Clip. Delete in its entirety

PAGE 25

* 4.4.2.14: Cap, Short Circuit, Delete in its entirety

PAGE 26

4.4.2.15: Major 102, under AQL column, delete ".40%' and substitute “100%”. Delete “1/”.

Delete Note 1 in its entirety.

Delete major defects 101, 103, 104 and minor defect 201 in their entirety.

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PAGE 27

4.4.2.16: Delete in its entirety,

PAGE 28

4.4.2.l7: Delete major defect 101 in its entirety.

Add “104 Match not completely coated, 100%, 3.2, Visual”.

PAGE 31

* 4.4.2.20: Delete in its entirety.

PAGE 32

4.4.2.21: Delete major 102 in its entirety,

PAGE 36A

* Add new page 36A included.

PAGE 36B

* Add new page 36B included.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.25	**TITLE** Plug, Electrical Assembly (Alternate)	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 12944400
			NEXT HIGHER ASSEMBLY 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined				
Major					
101	Thickness of body, max.		0.40%	3.2	Gage
102	Width under head (Datum A)		0.40%	3.2	Gage
103	Depth under head (Datum B)		0.40%	3.2	Gage
104	Pins missing or damaged		0.40%	3.2	Visual
105	True position of pins		0.40%	3.2	Gage
106	Pin insert shoulders above flush, max.		0.40%	3.2	Gage
107	Pull test, pins to plug body	30	0-1	3.2	Gage
108	Pull Test, cable to plug body	30	0-1	3.2	Gage
109	Diameter of pin head		0.40%	3.2	Gage
110	Length of pin head, min.		0.40%	3.2	Gage
111	Width of slots		0.40%	3.2	Gage
Minor					
201	Length of cable		0.65%	3.2	Gage
202	Shortening clip strap improperly assembled		0.65%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.26	**TITLE** Igniter, Electrical Assembly (Alternate)	**SHEET** 1 **OF** 1	**DRAWING NUMBER** **12972336**
			NEXT HIGHER ASSEMBLY 9342134

CATEGORY	**EXAMINATION OR TEST**	**NO. OF SAMPLE UNITS**	**AQL OR 100%**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE/ INSPECTION METHOD**
Critical	None defined				
Major					
101	Match insecure or improperly assembled (soldered)		0.40%	3.2	Visual/Manual
102	Match not completely coated		100%	3.2	Visual
103	Electrical resistance test		100%	3.2	Ohmmeter
Minor					
201	Poor workmanship		1.00%	3.8	Visual

AMSMC Form 1570b, 1 Jul 89

Replaces 1570a, 1 Feb 85, which may not be used.

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PAGE 37

4.4.5.1, line 7: Delete “The lot shall be rejected in one (1) or more…” and substitute “The lot shall be rejected if two (2) or more …”.

4.4.5.2, line 7: Delete “The lot shall be rejected in one (2) or more…” and substitute “The lot shall be rejected if two (3) or more …”.

PAGE 39

4.5.2.1 Charge composition. Line 3, delete “Method 102.1” and substitute “Method 102.2.1”.

PAGE 40

4.5.3.6: Delete in its entirety.

PAGE 41

4.5.6, line 2: Delete “(70 ± 5°F)

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